



Self-Learning and Methodology Documentation



Objective

The goal was to analyze Google Play Store app data to identify potential app categories for future app development. The focus was on finding categories likely to achieve high user engagement and higher install counts, thereby maximizing ad revenue.



Tools & Libraries Used

- **Python:** Main programming language for data analysis.
 - **Pandas:** For data loading, cleaning, and analysis.
 - **Google Colab / Jupyter Notebook:** Interactive environment for code execution and exploration.
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Methodology



Data Loading and Inspection

- The dataset was loaded using `pd.read_csv()`.
- Initial exploration was done with `df.head()` and `df.info()` to understand the structure and identify necessary cleaning steps.



Duplicate Records Handling

- Duplicate rows were identified using `df.duplicated()` and counted with `sum()`.
- Duplicates based on the `App` column were removed using `drop_duplicates(subset=['App'], keep='first')`, ensuring only irrelevant or redundant records were deleted.



Column Cleaning

- The `Installs` column was cleaned by removing characters such as "," and "+", then converted to integer for numerical analysis.
- Columns were renamed where needed, such as changing `Size` to `Size_in_App` for better clarity.



Category-Wise Analysis

- Used `groupby('Category')` combined with `mean()` to calculate the average installs per category.
- Categories were sorted by average installs to identify those with high engagement.

Category Selection

- Categories dominated by strong market leaders (e.g., Communication) were excluded to avoid heavy competition.
 - Remaining categories with high average installs were shortlisted for new app ideas.
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Concepts Learned

- Strategic duplicate removal using Pandas while preserving valid data.
 - Column transformation and data type conversion for analysis readiness.
 - Group-wise aggregation and strategic filtering based on business criteria.
 - Key Pandas functions: `drop_duplicates`, `groupby`, `mean`, `replace`, `apply`, and column renaming.
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Insights Gained

- High average installs indicate strong engagement but do not always imply good market entry due to dominant players.
 - Moderately popular categories may offer higher chances for successful entry and revenue growth.
 - Clean, accurate data is crucial before performing any business analysis to avoid misleading conclusions.
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External References

- [Pandas Documentation](#)
 - Python official documentation
 - Google Colab official guide and tutorials
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Final Deliverables

- Cleaned dataset without duplicate records.
- Report on average installs per category.
- Recommended app categories for future development.
- Self-learning and methodology documentation (this document).